# StegaGuard: A Controlled Robustness Study of Feature-Based Image Steganalysis under Algorithm and Payload Shift

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**Abstract.** Steganalysis detectors are trained on known embedding algorithms and payloads, but an adversary can change both. StegaGuard is a controlled robustness study of two existing feature-based detectors: Residual-LR (76 residual statistics, logistic regression) and SRM+Ensemble (34,671 Spatial Rich Model features, Fisher linear discriminant ensemble). We split 600 BOSSbase 1.01 covers, resized to 256 × 256, into 360 training, 120 validation and 120 test covers before embedding them with WOW, S-UNIWARD and HILL at 0.1 and 0.4 bpp, so every stego variant stays in the partition of its source cover. Three training protocols of 360 cover–stego pairs were run with three seeds, and paired differences were assessed by a bootstrap over test covers. Trained on WOW at 0.4 bpp, SRM+Ensemble reached an AUC of 0.714 on matched WOW and 0.645 on unseen HILL (ΔAUC = −0.070, 95% CI [−0.093, −0.046]). Adding 0.1 bpp variants to a WOW/S-UNIWARD training pool gave no reliable change on unseen HILL (ΔAUC = −0.022, 95% CI [−0.050, +0.005]). AUC at 0.1 bpp was lower than at 0.4 bpp in all 18 comparisons (ΔAUC −0.022 to −0.183). At thresholds set for a validation false-positive rate (FPR) of at most 1%, mean recall was 0.062 and 0.080 while the test FPR was 0.045 and 0.034, so neither detector operated reliably at a 1% false-alarm budget on this resized BOSSbase subset.

**Keywords:** Image Steganalysis, Robustness Evaluation, Algorithm Mismatch, Payload Mismatch, Spatial Rich Model, Low False-Positive Rate.

## 1 Introduction

Image steganography hides a message by making small changes to a cover image. Current spatial-domain schemes such as WOW, S-UNIWARD and HILL assign a distortion cost to every pixel and use syndrome-trellis coding, or a simulation of it, to place most changes in textured regions that are hard to model [1–4]. Steganalysis detectors either train a classifier on statistics of noise residuals, such as SPAM or the Spatial Rich Model (SRM) [5–7], or learn the residual filters with a convolutional network such as Yedroudj-Net or SRNet [8, 9].

In a security setting the detector does not choose its test data: a defender trains on a few algorithms at known payloads, while an adversary may use an unseen algorithm or a shorter message. The operating point matters as well: a screening system inspects

many innocent images, so what counts is the share of stego images caught when the false-positive rate (FPR) is held near 1%, not only the area under the ROC curve (AUC).

Both issues are known, but they are usually studied one at a time. Work on cover-source mismatch changes where the images come from [10–12], work on stego-source mismatch changes the embedding method [13, 14], and papers that introduce detectors mostly train and test on the same algorithm and payload [6, 8, 9]. Two design choices can also bias robustness claims: splitting after embedding lets variants of one cover fall on both sides and inflate test scores, and a detector that is weak in the matched condition says little about the shift itself.

The gap has two parts. For the detector, it is unclear how much a standard SRM+Ensemble loses on a small, resized BOSSbase subset when the algorithm or payload changes, and whether mixing algorithms and payloads in training reduces the loss. For the evaluation, few shift studies combine a source-cover split fixed before embedding, a matched-condition reference, paired intervals on identical test covers and an explicit low-FPR operating point. StegaGuard (Fig. 1) addresses the second part to answer the first; it evaluates two existing detectors and proposes no new detection algorithm. Three questions guide the study:

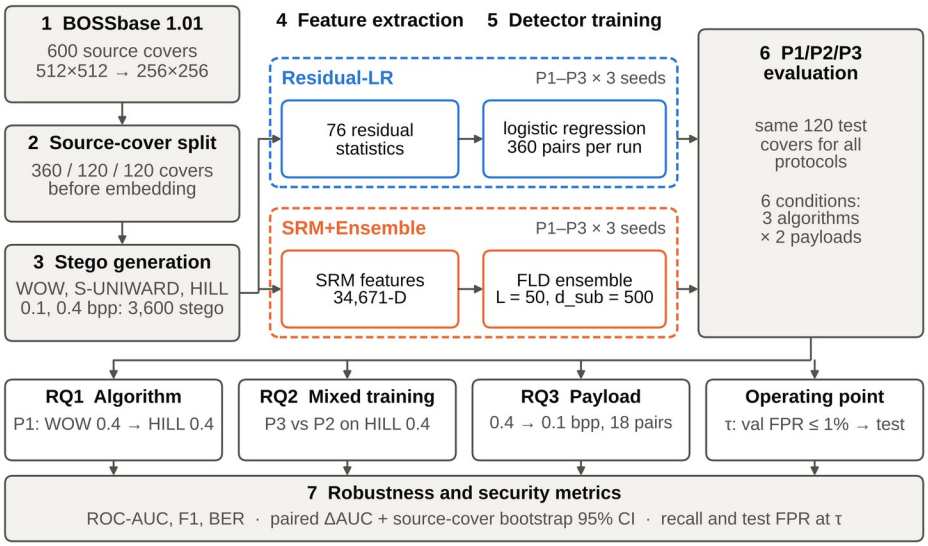



**Fig. 1.** StegaGuard evaluation pipeline. Both detectors are trained on the same splits, protocols and seeds, and every model is scored on the same 120 test covers.

- **RQ1 (unseen algorithm).** Under protocol P1, which trains on WOW at 0.4 bpp, how does detection of unseen HILL at 0.4 bpp compare with detection of matched WOW at 0.4 bpp on the same test covers?
- **RQ2 (mixed training).** With training size fixed at 360 cover–stego pairs, does adding 0.1 bpp variants to a WOW/S-UNIWARD training pool (P3 versus P2) improve detection of unseen HILL at 0.4 bpp?

- **RQ3 (payload shift).** For each detector, protocol and algorithm, how much does detection change when the payload falls from 0.4 to 0.1 bpp?

**Contributions.** (i) A controlled evaluation in which both detectors share covers, training size, seeds and test images, so that differences between protocols P1–P3 trace back to the training pool. (ii) Source-cover leakage prevention: the split is fixed before embedding and audited against a hashed manifest. (iii) A robustness assessment of algorithm, mixed-training and payload shift with paired ΔAUC and bootstrap intervals over test covers. (iv) A security operating-point analysis that separates the validation-selected threshold from the FPR observed on test covers.

## 2 Related Work

### 2.1 Image Steganalysis Techniques

Content-adaptive embedding places changes where the cover is noisy. WOW derives its costs from directional wavelet residuals, S-UNIWARD from the relative change of a wavelet decomposition, and HILL from a high-pass residual smoothed by low-pass filters [2–4]. Because the three cost functions favor different structures, a detector trained on one algorithm meets a partly different change pattern when tested on another. Feature-based steganalysis models the noise residual: SPAM uses Markov transition probabilities of pixel differences [5], and SRM gathers co-occurrences of many linear and non-linear residuals into 34,671 features [6], usually classified by the FLD ensemble, which handles this dimension through random subspaces and bagging [7]. CNN detectors such as Yedroudj-Net and SRNet learn the filters and report lower error than SRM with the ensemble on BOSSbase-derived data, at a higher training cost [8, 9].

### 2.2 Robustness under Algorithm and Payload Shifts

Three kinds of train–test shift should be kept apart. *Algorithm shift*, or stego-source mismatch, occurs when test stego images come from an embedding method absent from training. Butora and Fridrich trained SRNet on three of WOW, S-UNIWARD, HILL and MiPOD at 0.4 bpp and tested on the fourth; the largest loss occurred when HILL was held out, while training on several algorithms cost little against dedicated detectors [13]. Recent work adapts to algorithm and cover-source mismatch with unlabeled target data [14]. *Payload shift* occurs when the message length differs between training and test. Detection error grows as the payload falls; SRNet built its low-payload detectors by curriculum training across payloads [9], and knowledge distillation has been proposed for test payloads unseen in training [15]. *Cover-source shift* arises when test covers come from other cameras or processing pipelines and can raise the error even when algorithm and payload match [10–12]. This study varies the algorithm and the payload and holds the cover source fixed.

Robustness also depends on the operating point. The ALASKA challenge ranked detectors by the missed-detection rate at 5% false alarms (MD5) [16], and the FLD ensemble has been extended to meet a prescribed false-alarm probability [17]. Butora

and Bas showed that two detectors with almost equal error can differ by about ten points in detection rate at an FPR of $10^{-3}$ [18].

### 2.3 Research Gap and Comparative Analysis

Table 1 compares representative studies; "Not reported" means the paper does not report the item.

**Table 1.** Comparison with representative steganalysis studies.

| Study | Data | Detector | Protocol and metric | Shift | Low-FPR metric |
|---|---|---|---|---|---|
| SRM [6] | BOSSbase 0.92, 512 × 512 | SRM + FLD ensemble | Matched algorithm and payload; $P_E$ | None | Not reported |
| Yedroudj-Net [8] | BOSSbase 1.01, resized to 256 × 256 | CNN | WOW, S-UNIWARD 0.2/0.4 bpp; matched; $P_E$ | None | Not reported |
| SRNet [9] | BOSSbase 1.01 + BOWS2, 256 × 256 | Residual CNN | 0.1–0.5 bpp, curriculum over payloads; $P_E$ | None | Not reported |
| CSM [10] | Several cover sources, spatial and JPEG | Rich features + FLD ensemble | Train and test on different sources | Cover source | Not reported |
| Butora & Fridrich [13] | BOSSbase 1.01 + BOWS2, 256 × 256 | SRNet variants | Leave one of 4 algorithms out, 0.4 bpp; $P_{MD}$ | Algorithm | No fixed FPR |
| ALASKA [16] | JPEG images with diverse processing | Challenge entries | Challenge ranking | Cover source | MD5 (5% FPR) |
| MODE [18] | BOSSbase, 512 × 512 | Pretrained SRNet | HILL 0.3 bpp; $P_E$, detection at fixed FPR | None | FPR $10^{-3}$–$10^{-1}$ |
| This study | BOSSbase 1.01, 600 covers, 256 × 256 | Residual-LR; SRM + FLD ensemble | Split before embedding; paired bootstrap | Algorithm, payload | Recall, test FPR at 1% val. FPR |

Detector papers evaluate the matched condition, shift studies mostly concern the cover source or use large CNN training sets [13], and low-FPR operation is studied apart from shift [18]. To our knowledge, no study measures algorithm and payload shift for feature-based detectors under a split fixed before embedding, with paired intervals on identical test covers, while also reporting the FPR actually reached at a validation-selected threshold. StegaGuard addresses this gap.

## 3 Methodology

Fig. 1 summarizes the pipeline. Stages 1–3 build a leakage-controlled dataset, stages 4–5 train the two detectors in parallel under protocols P1–P3, and stages 6–7 evaluate every trained model on one fixed set of test covers.

### 3.1 Dataset Preparation and Leakage Prevention

BOSSbase 1.01 contains 10,000 grayscale images of 512 × 512 pixels [19]. We drew 600 of them (6%) with the fixed seed 20260918 and downsampled each to 256 × 256

before embedding, as several CNN studies do [8, 9]. Resizing alters cover noise statistics, so our numbers are not comparable with results at 512 × 512.

Leakage is prevented by the order of operations. The 600 source covers were first assigned to training (360), validation (120) and test (120). Only then was each cover embedded with WOW [2], S-UNIWARD [3] and HILL [4] at 0.1 and 0.4 bpp, giving six variants per cover: 600 stego images per algorithm–payload condition, 3,600 in total and 4,200 PNG files with the covers. Training, validation and test hold 2,160, 720 and 720 stego images (360, 120 and 120 per condition). Each stego file inherits the cover_id and split of its source, so a cover and its six variants always share a partition. The manifest records cover_id, split, algorithm, payload, embedding seed and SHA-256 hash; an audit confirmed that every cover_id lies in exactly one partition with exactly six variants and that no cover_id overlaps (manifest SHA-256 13c01ca9…2799e8c).

### 3.2 Feature Extraction and Detectors

Both detectors output a real-valued score, higher for stego, that is used for every metric; Table 2 summarizes their configuration. Residual-LR, the lightweight baseline, describes each image by 76 statistics of its noise residuals and classifies them with scikit-learn logistic regression [20]. SRM+Ensemble, the established reference, extracts the 106 SRM submodels (34,671 features) [6] with the sealwatch library [21] and trains the Fisher linear discriminant (FLD) ensemble of [7] with $L = 50$ base learners and random feature subspaces of dimension $d_{sub} = 500$. Both detectors keep the same settings in every protocol and seed, so differences between protocols come from the training data alone.

**Table 2.** Detector configuration.

| Setting | Residual-LR | SRM+Ensemble |
|---|---|---|
| Features | 76 residual statistics | SRM: 106 submodels, 34,671 dimensions |
| Classifier | logistic regression (scikit-learn) | FLD ensemble, $L = 50$, $d_{sub} = 500$ |
| Training data per run | 360 cover–stego pairs | 360 cover–stego pairs |
| Seed-dependent parts | variant sampling (P2, P3) | variant sampling (P2, P3); random parts of ensemble training |
| Output | real-valued score, higher for stego | real-valued score, higher for stego |

### 3.3 Experimental Protocols

The protocols in Table 3 differ only in the stego variant attached to each of the 360 training covers. For every training cover and seed, one variant is sampled from the protocol's pool, so each run sees 360 cover–stego pairs built from the same 360 covers; P1 involves no sampling. Validation images serve only to set thresholds. HILL never enters training or validation, and S-UNIWARD is unseen under P1. Seeds 42, 43 and 44 control the variant draws and the random parts of model fitting.

**Table 3.** Training protocols. All share the same 360 training, 120 validation and 120 test covers; every model is tested on all six conditions (3 algorithms × 2 payloads).

| Protocol | Training stego pool | Variants in pool | Role |
|---|---|---|---|
| P1 | WOW 0.4 bpp | 1 | RQ1 (WOW 0.4 matched, HILL 0.4 unseen); RQ3 |
| P2 | WOW and S-UNIWARD at 0.4 bpp | 2 | RQ2 reference; RQ3 |
| P3 | WOW and S-UNIWARD at 0.1 and 0.4 bpp | 4 | RQ2 (P3 − P2 on HILL 0.4); RQ3 |

RQ1 uses P1 because only there is WOW 0.4 the sole training condition, so WOW 0.4 is matched and HILL 0.4 unseen for the same model. P2 and P3 differ only in the 0.1 bpp variants, which RQ2 isolates. In every comparison both conditions are scored on the same 120 test covers.

### 3.4 Evaluation Metrics

For each run and test condition a detector produces 240 scores (120 test covers and their 120 stego variants), from which ROC-AUC is computed. Two thresholds are fixed from validation scores alone, never from test labels. The first, $\tau_{BER}$, minimizes the balanced error rate BER = $(P_{FA} + P_{MD})/2$ on the validation scores, where $P_{FA}$ and $P_{MD}$ are the false-alarm and missed-detection rates; F1 = 2·Prec·Rec/(Prec + Rec) and BER are reported on test at this threshold. The second threshold defines the security operating point. With s(c) the score of cover c and V the set of 120 validation covers,

$$\tau_{low} = min\left\{\tau : \left|\left\{c \in V : s(c) \geq \tau\right\}\right| \leq 0.01 \cdot |V|\right\} \quad (1)$$

With |V| = 120, the constraint allows at most one validation false positive (0.83%). At $\tau_{low}$ we report recall, the share of the 120 test stego images with $s \geq \tau_{low}$, and the observed test FPR, the share of the 120 test covers with $s \geq \tau_{low}$. The first is the quantity of security interest; the second shows whether the 1% target transfers from validation to test, which Eq. (1) does not guarantee.

### 3.5 Statistical Analysis and Reproducibility

Each comparison is a paired difference ΔAUC = AUC(shifted) − AUC(reference): HILL 0.4 minus WOW 0.4 under P1 (RQ1), P3 minus P2 on HILL 0.4 (RQ2), and 0.1 minus 0.4 bpp for each detector, protocol and algorithm (RQ3, 18 comparisons). Negative values mean a loss under the shift. The point estimate is the mean of the three per-seed differences, which equals the difference of the seed-mean AUCs. Its 95% interval comes from a paired bootstrap in which the resampling unit is the source test cover: each replicate draws 120 test cover IDs with replacement, and a drawn cover brings its cover score together with its stego scores under both compared conditions, so the pairing is preserved. Individual stego images are never resampled on their own.

Seed standard deviations (at most 0.028, Supplementary Table S1) describe training randomness on fixed data; the bootstrap describes the sampling of test covers given the

trained models. Neither covers another choice of covers or split, so the intervals are conditional on this partition. A direction is called supported only when its interval excludes zero; an interval containing zero is inconclusive, not evidence of equivalence. The 18 RQ3 intervals are not adjusted for multiplicity. Every run stores its per-image prediction scores, and the manifest stores the embedding seed and SHA-256 hash of every image, so all tables can be recomputed from the saved files.

# 4 Results and Discussion

## 4.1 Overall Detection Performance

Table 4 gives the ROC-AUC for every detector, protocol and test condition; F1, BER, recall and seed standard deviations are in Supplementary Table S1. All cells share the 120 test covers, so they are paired observations, not independent experiments. Residual-LR stays close to chance, with AUC of 0.500–0.570 and BER of at least 0.438. SRM+Ensemble separates covers from stego at 0.4 bpp (AUC 0.632–0.717, BER 0.347–0.400) but hardly at 0.1 bpp (0.517–0.550). As Residual-LR fails even in the matched condition, SRM+Ensemble carries the main evidence for the research questions.

**Table 4.** ROC-AUC (mean over seeds 42–44) on the 120 test covers at 0.4 and 0.1 bpp. Bold: the matched condition of P1. HILL is unseen under all protocols; S-UNIWARD is unseen under P1.

| Detector | P | WOW | | S-UNIWARD | | HILL | |
|---|---|---|---|---|---|---|---|
| | | **0.4** | **0.1** | **0.4** | **0.1** | **0.4** | **0.1** |
| Residual-LR | P1 | **0.546** | 0.510 | 0.565 | 0.505 | 0.570 | 0.514 |
| | P2 | 0.534 | 0.500 | 0.549 | 0.504 | 0.551 | 0.505 |
| | P3 | 0.523 | 0.501 | 0.555 | 0.502 | 0.546 | 0.511 |
| SRM+Ensemble | P1 | **0.714** | 0.539 | 0.689 | 0.532 | 0.645 | 0.525 |
| | P2 | 0.713 | 0.550 | 0.717 | 0.534 | 0.653 | 0.534 |
| | P3 | 0.686 | 0.546 | 0.677 | 0.528 | 0.632 | 0.517 |

## 4.2 RQ1: Unseen Embedding Algorithm

Under P1, SRM+Ensemble falls from 0.714 on matched WOW 0.4 to 0.645 on unseen HILL 0.4 (Table 5); the interval of the paired ΔAUC excludes zero, and BER rises from 0.347 to 0.386. For Residual-LR the difference is +0.024 [−0.007, +0.056], but with a matched AUC of 0.546 there is no matched signal against which a shift could be measured.

This loss is not a pure effect of algorithm mismatch. HILL was designed to be harder to detect than WOW with rich-model features [4], so part of the gap may reflect its intrinsic difficulty; the P1 ordering WOW 0.714 > S-UNIWARD 0.689 > HILL 0.645 fits both explanations. Separating them needs a matched HILL reference (trained and tested on HILL), which this study did not run.

**Answer to RQ1.** On this subset, SRM+Ensemble trained on WOW at 0.4 bpp detects unseen HILL at 0.4 bpp less well than matched WOW on the same test covers (ΔAUC = −0.070, 95% CI [−0.093, −0.046]). This is a measured degradation under the tested shift; it does not establish robustness, or a general lack of it, for other algorithms, payloads or cover sources.

**Table 5.** Paired comparisons for RQ1 and RQ2 on the same 120 test covers. ΔAUC is computed from unrounded scores and may differ by 0.001 from the difference of the rounded AUCs.

| Detector | Reference AUC | Shifted AUC | ΔAUC | 95% CI | Reading |
|---|---|---|---|---|---|
| *RQ1: matched WOW 0.4 → unseen HILL 0.4, both under P1* | | | | | |
| Residual-LR | 0.546 | 0.570 | +0.024 | [−0.007, +0.056] | No matched signal |
| SRM+Ensemble | 0.714 | 0.645 | −0.070 | [−0.093, −0.046] | Loss, CI excludes 0 |
| *RQ2: P2 → P3, both tested on unseen HILL 0.4* | | | | | |
| Residual-LR | 0.551 | 0.546 | −0.005 | [−0.020, +0.007] | Inconclusive |
| SRM+Ensemble | 0.653 | 0.632 | −0.022 | [−0.050, +0.005] | Inconclusive |

### 4.3 RQ2: Mixed Training

P3 adds 0.1 bpp WOW and S-UNIWARD variants to the P2 pool at the same 360 pairs, so fewer training pairs carry the stronger 0.4 bpp signal. On unseen HILL 0.4, SRM+Ensemble scores 0.653 under P2 and 0.632 under P3 (ΔAUC = −0.022, 95% CI [−0.050, +0.005]); for Residual-LR the difference is −0.005 [−0.020, +0.007]. Both intervals include zero, so the data show neither an improvement nor equivalence of the two protocols. For SRM+Ensemble the point estimates also fall on matched 0.4 bpp conditions (WOW 0.713 → 0.686, S-UNIWARD 0.717 → 0.677), and AUC at 0.1 bpp does not rise although 0.1 bpp is now trained on. One untested explanation is that, at fixed size, weak 0.1 bpp examples dilute the 0.4 bpp signal.

**Answer to RQ2.** No reliable improvement was observed under the evaluated mixed-training protocol. This holds for WOW and S-UNIWARD as training algorithms, two payloads, 360 training pairs and this dataset; larger or differently weighted mixed training sets remain untested.

### 4.4 RQ3: Payload Shift

Fig. 2 shows ΔAUC between 0.1 and 0.4 bpp for all 18 comparisons, and Table 6 summarizes them per detector and protocol; Supplementary Table S2 lists every interval. All 18 differences are negative and all 18 intervals lie below zero. For SRM+Ensemble the losses range from −0.115 to −0.183 with upper bounds no higher than −0.088; for Residual-LR they are smaller (−0.022 to −0.060) and one interval ends at −0.001 (P3 WOW). As the intervals are unadjusted and share covers and models, the common sign is not 18 independent confirmations, and the weakest Residual-LR intervals are not strong evidence on their own.

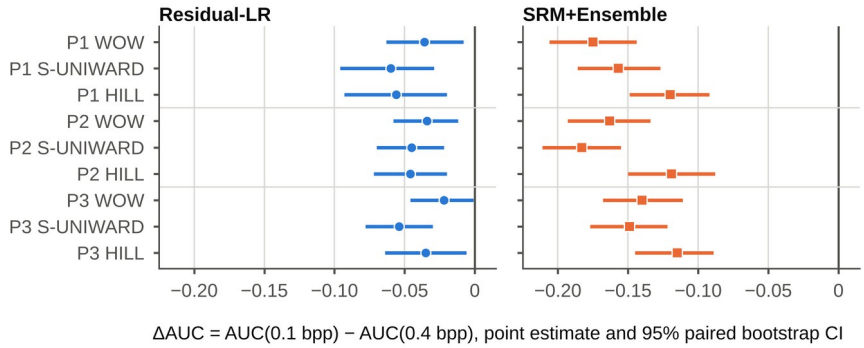



**Fig. 2.** Payload shift: ΔAUC = AUC(0.1 bpp) − AUC(0.4 bpp) with 95% paired bootstrap intervals. Under P1 and P2 the 0.1 bpp payload is absent from training; under P3 it is included.

**Table 6.** RQ3 summary over the three algorithms, ΔAUC = AUC(0.1 bpp) − AUC(0.4 bpp).

| Detector | P | AUC at 0.1 bpp | ΔAUC range | Mean ΔAUC | Highest CI upper bound | CIs < 0 |
|---|---|---|---|---|---|---|
| Residual-LR | P1 | 0.505–0.514 | −0.060 to −0.036 | −0.051 | −0.008 | 3/3 |
| | P2 | 0.500–0.505 | −0.046 to −0.034 | −0.042 | −0.012 | 3/3 |
| | P3 | 0.501–0.511 | −0.054 to −0.022 | −0.037 | −0.001 | 3/3 |
| SRM+Ensemble | P1 | 0.525–0.539 | −0.175 to −0.120 | −0.151 | −0.092 | 3/3 |
| | P2 | 0.534–0.550 | −0.183 to −0.119 | −0.155 | −0.088 | 3/3 |
| | P3 | 0.517–0.546 | −0.149 to −0.115 | −0.135 | −0.089 | 3/3 |

**Answer to RQ3.** Lowering the payload from 0.4 to 0.1 bpp reduced AUC in every tested combination, and for SRM+Ensemble (0.517–0.550 at 0.1 bpp) it removed most of the usable signal. The loss persists under P3, whose pool contains 0.1 bpp examples, so at 360 pairs training on the lower payload did not offset the weaker signal. The finding is limited to the three algorithms, two payloads and image size studied.

### 4.5 Security Operating Point at Low FPR

Table 7 reports the operating point of Eq. (1). The threshold allows at most one false positive among the 120 validation covers, yet on test covers every detector–protocol cell exceeds the 1% target, with observed FPRs of 0.022–0.053. Residual-LR under P1 is deterministic and gives 5 false positives in 120 test covers per run; the exact 95% Clopper–Pearson interval for 5/120, [1.4%, 9.5%], excludes 1%. A likely reason is that $\tau_{low}$ depends on the one or two highest validation cover scores, an extreme order statistic that varies strongly between samples of 120.

**Table 7.** Operating point at $\tau_{low}$ (validation FPR ≤ 1%). False positives are summed over seeds 42–44 (3 × 120 test covers); recall is the mean and range over the six test conditions.

| Detector | P | Test false positives | Observed test FPR | Mean recall | Recall range |
|---|---|---|---|---|---|
| Residual-LR | P1 | 15/360 | 0.042 | 0.049 | 0.042–0.058 |
| | P2 | 15/360 | 0.042 | 0.070 | 0.053–0.097 |
| | P3 | 19/360 | 0.053 | 0.068 | 0.056–0.092 |
| SRM+Ensemble | P1 | 18/360 | 0.050 | 0.097 | 0.053–0.189 |
| | P2 | 11/360 | 0.031 | 0.069 | 0.025–0.156 |
| | P3 | 8/360 | 0.022 | 0.075 | 0.019–0.147 |
| *Pooled: Residual-LR 49/1,080 (FPR 0.045, mean recall 0.062); SRM+Ensemble 37/1,080 (FPR 0.034, mean recall 0.080)* | | | | | |

Even the best cell, SRM+Ensemble on matched WOW 0.4 under P1, reached a recall of only 0.189 at a test FPR of 0.050. Each detector has 54 evaluation rows (3 protocols × 3 seeds × 6 conditions); in none of them were recall and FPR both zero, so the detectors were never completely silent. The rows hold 6,480 stego scores per detector, but these come from 720 distinct test stego images of 120 source covers and are repeated measurements, not 6,480 independent samples. The detectors thus did not demonstrate reliable operation at FPR ≤ 1%, and the results do not support deployment in an operational screening system.

### 4.6 Discussion and Security Implications

*Observed evidence.* SRM+Ensemble lost AUC when moving from matched WOW to unseen HILL; this mirrors the largest loss for held-out HILL reported for SRNet [13], obtained there with about 28,000 training images instead of 720. Mixing payloads at a fixed training size did not help on HILL; Butora and Fridrich likewise found in preliminary tests that a random mixture of payloads trained no better than a single mid payload [13]. Both detectors were far weaker at 0.1 bpp, matching the payload dependence reported for SRM and SRNet [6, 9]. The observed test FPR exceeded the validation target, which supports the warning that low-FPR estimates from small test sets are noisy [18]. *Possible explanations*, not tested here, are the higher intrinsic security of HILL [4], the small training set and the resizing to 256 × 256.

For a defender, ranking quality and operational usefulness diverge: an AUC of 0.71 still meant missing over 80% of stego images at a threshold meant to keep false alarms near 1%. For an adversary, switching to HILL or to 0.1 bpp was enough to weaken detection. The main limits are the dataset size (600 covers, one split, so the intervals are conditional on that partition), the resizing to 256 × 256, detector strength (two feature-based detectors, one near chance, and no matched HILL reference) and generalization (one cover source, and a 1% threshold calibrated on 120 covers).

## 5 Conclusion

With the source-cover split fixed before embedding, the three research questions received different answers. For RQ1, SRM+Ensemble trained on WOW at 0.4 bpp detected unseen HILL less well than matched WOW ($\Delta$AUC = −0.070, 95% CI [−0.093, −0.046]), although this loss cannot be separated from HILL being harder to detect. For RQ2, adding 0.1 bpp variants to the training pool gave no reliable improvement on unseen HILL. For RQ3, all 18 comparisons showed lower AUC at 0.1 than at 0.4 bpp, and SRM+Ensemble came close to chance at the lower payload.

These conclusions are limited to 600 BOSSbase covers resized to 256 × 256 and to two feature-based detectors, one of them close to chance. The 1% operating point was calibrated on 120 validation covers and was not met on test covers, and no cross-source data were evaluated.

Future work will use the full BOSSbase at its original resolution, add stronger detectors such as SRNet under the same split, include a matched HILL reference and test cover-source shift. StegaGuard is an evaluation protocol rather than a new detector, and the results do not indicate readiness for deployment.

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